# Entanglement at the Institute for Quantum Information




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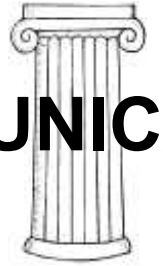
# «Entanglement»

VIEW

Entanglement as the basic proving ground for understanding what makes quantum information science **different**



**LOCALITY**

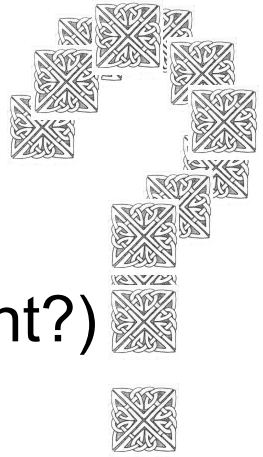


**COMMUNICATION**

Entanglement as the answer to all our questions?
…probably asking too much (dynamics?!)

Entanglement as a quintessential quantum phenomenon
…use as test bed, jumping off point, inspiration!

# Entanglement Questions



When is a mixed state entangled?

(when has an experiment generated entanglement?)

What states admit local classical theories?

(when does nature act like a classical computer?)

What is communication cost of simulating quantum correlations?

(when can quantum computers lessen communication?)

What is the entanglement cost of a mixed state?

How can we develop a theory of multipartite entanglement?

What is the role of entanglement in quantum critical phenomena?

(what role does entanglement play in many body ground states?)

# So You Think You're Entangled?

| Separable | Entangled |
|---|---|
| **pure states** | |
| $\|\psi\rangle = \|\mu\rangle_A \otimes \|\nu\rangle_B$ | $\|\psi\rangle = \sum_i \sqrt{p_i}\|\mu_i\rangle_A \otimes \|\nu_i\rangle_B \neq \|\mu\rangle_A \otimes \|\nu\rangle_B$ |
| **mixed states** | |
| $\boldsymbol{\rho} = \sum_i p_i \|\psi_i\rangle\langle\psi_i\|_A \otimes \|\eta_i\rangle\langle\eta_i\|_B$ | $\boldsymbol{\rho} \neq \sum_i p_i \|\psi_i\rangle\langle\psi_i\|_A \otimes \|\eta_i\rangle\langle\eta_i\|_B$ |

How do we determine if a mixed state is entangled?

Experimentally how do we prove entanglement?

# Separability Criteria

Peres-Horodecki Partial Transpose Separability Criteria:

$$\boldsymbol{\rho}_{AB} \text{ separable} \Rightarrow \boldsymbol{\rho}_{AB}^{T_A} \geq 0$$

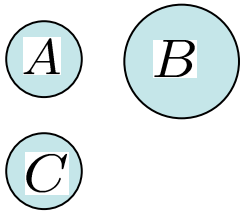
$$\rho_{ik,jl} = \langle i|_A \otimes \langle k|_B \boldsymbol{\rho}_{AB} |j\rangle_A \otimes |l\rangle_B \qquad \rho_{ik,jl}^{T_A} = \rho_{jk,il}$$

partial transpose on A

$$\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|_A \otimes |\eta_i\rangle\langle\eta_i|_B\right)^{T_A} = \sum_i p_i |\psi_i^*\rangle\langle\psi_i^*|_A \otimes |\eta_i\rangle\langle\eta_i|_B$$

Separable state extension on copy of A:

$$\sum_i p_i |\psi_i\rangle\langle\psi_i|_A \otimes |\eta_i\rangle\langle\eta_i|_B \otimes |\psi_i\rangle\langle\psi_i|_C$$




$$\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_A$$

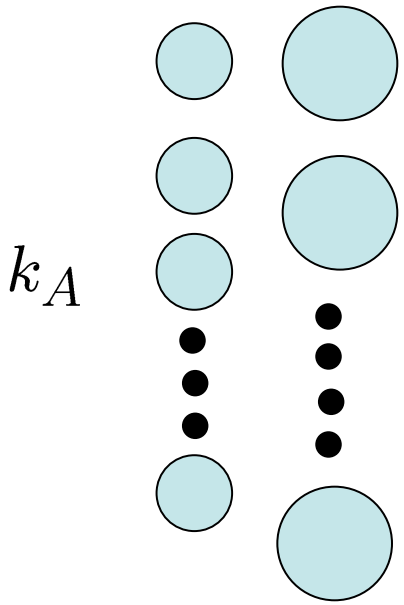
$\boldsymbol{\rho}_{AB}$ separable $\Rightarrow$ exists $\boldsymbol{\rho}_{ABC}$ which is

symmetric: $(AC)\boldsymbol{\rho}_{ABC}(AC) = \boldsymbol{\rho}_{ABC}$

$$\boldsymbol{\rho}_{ABC}^{T_A} \geq 0, \boldsymbol{\rho}_{ABC}^{T_B} \geq 0$$

# A Family of Separability Criteria

"Distinguishing Separable and Entangled States," Doherty, Parrillo, Spedalieri, PRL, **88** 187904 (2002)




$\mathcal{H}_A^{\otimes k_A} \otimes \mathcal{H}_B^{\otimes k_B}$

If $\rho_{AB}$ is separable, then

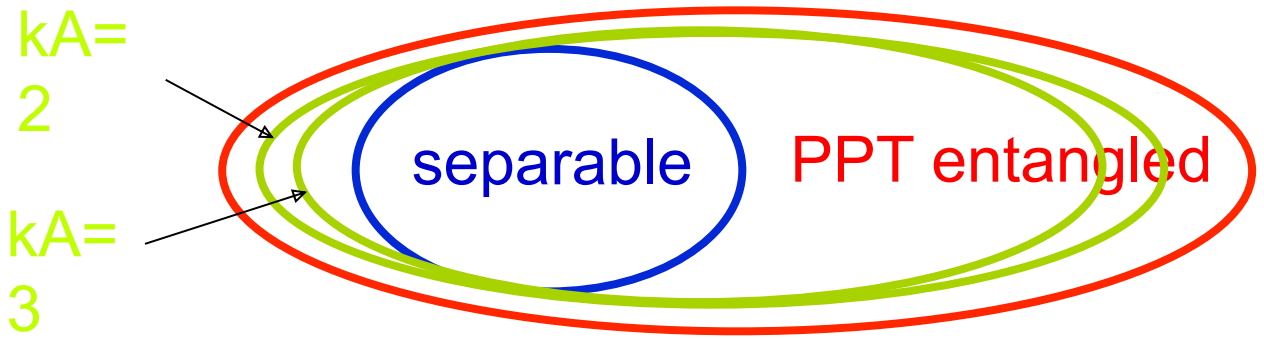
Exists a state $\tilde{\rho}$ on $\mathcal{H}_A^{\otimes k_A} \otimes \mathcal{H}_B^{\otimes k_B}$ such that

→ $Tr_{\neq AB}[\tilde{\rho}] = \rho_{AB}$

→ $\tilde{\rho}$ is symmetric under exchange of $\mathcal{H}_A$ (or $\mathcal{H}_B$) subsystem copies

→ $\tilde{\rho}^{T_S} \geq 0$ for all $S$

**Strictly stronger** separability criteria than Peres-Horodecki!



# Usefulness

- $Tr_{\neq AB}[\tilde{\rho}] = \rho_{AB}$
- $\tilde{\rho}$ is symmetric under exchange of $\mathcal{H}_A$ (or $\mathcal{H}_B$) subsystem copies
- $\tilde{\rho}^{T_S} \geq 0$ for all $S$

finding $\tilde{\rho}$ is a semidefinite program

(efficient search algorithms)

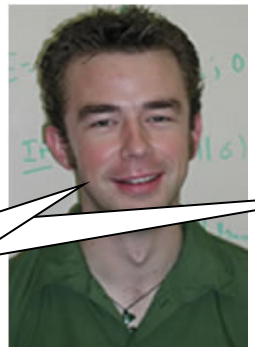
If criteria implies $\rho_{AB}$ entangled, semidefinite program gives

measure in lab to prove entanglement

entanglement witness $\mathbf{Z}$

$\mathrm{Tr}[\rho_{sep}\mathbf{Z}] \geq 0, \mathrm{Tr}[\rho_{AB}\mathbf{Z}] < 0$

**Constructive, efficient, complete criteria for mixed state separability!**

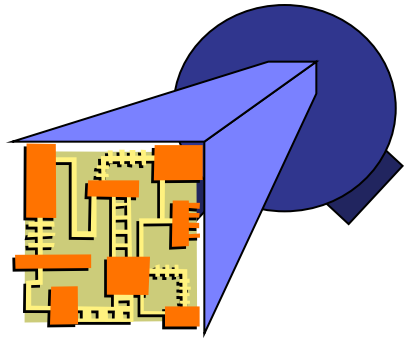


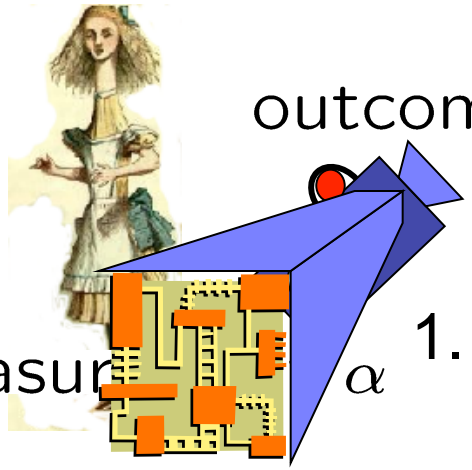
Doherty



Spedalieri

# Local Realist Theories





outcome $a$

outcome $b$



measur $\alpha$

1. Distribute quantur

measurement $\beta$

2. Parties select W measurement to perform

ALICE

BOB

3. Parties get measurement OUTCOMES

$Pr[a, b|\alpha, \beta] =$ proability outcome $a, b$ given $\alpha, \beta$

Can we explain $Pr[a, b|\alpha, \beta]$ via local classical hidden variables?

# Nonlocality of All Entangled States?



Bell's Theorem: Correlations produced by measurements on *some* entangled states can not be explained by local realistic theories.

What is the relationship between entanglement and nonlocality for mixed states?

For certain entangled mixed states, all measurements on a single copy of the entangled state can be explained by local realistic theories (Werner).

**General methods for constructing local realistic theories for quantum states?**



Doherty



Terhal

# Local Realistic Theories for Quantum States

"Local Hidden Variable Theories for Quantum States," Terhal, Doherty, quant-ph/0210053 (2002)

<u>weaker but more intuitive version:</u>

If there exists a state $\tilde{\boldsymbol{\rho}}$ on $\mathcal{H}_A^{\otimes k_A} \otimes \mathcal{H}_B^{\otimes k_B}$ such that

$Tr_{\neq AB}[\tilde{\boldsymbol{\rho}}] = \boldsymbol{\rho}_{AB}$

$\tilde{\boldsymbol{\rho}}$ is symmetric under exchange of $\mathcal{H}_A$ (or $\mathcal{H}_B$) subsystems

Then there is a local hidden variable theory for $k_A$ ($k_B$) different measurements on $A$ ($B$)

note: ANY set of valid quantum measurements!

$\mathbf{B} = \mathbf{A}_1(\mathbf{B}_1 + \mathbf{B}_2) + \mathbf{A}_2(\mathbf{B}_1 - \mathbf{B}_2)$ LHV: $\mathrm{Tr}\,[\boldsymbol{\rho}_{AB}\mathbf{B}] \leq 2$

$$\mathrm{Tr}\,[\boldsymbol{\rho}_{AB}\mathbf{B}] = \mathrm{Tr}\,[\tilde{\boldsymbol{\rho}}\mathbf{B}] = \mathrm{Tr}\,[\tilde{\boldsymbol{\rho}}\textstyle\sum_i \mathbf{B}_i] = \mathrm{Tr}\left[\tilde{\boldsymbol{\rho}}\textstyle\sum_i \tilde{\mathbf{B}}_i\right]$$

$\tilde{\mathbf{B}}_i$ commute!

# Capitalism for Correlations

Shared randomness
Communication
**Quantum correlations**
**Super-quantum correlations**



Shared randomness
No communication
**Classical correlations**

Bell inequalities

How much communication is necesary to simulate quantum correlations?

What are the COSTS of producing quantum correlations?



# Simulating Quantum Correlations



1. Initial unlimited classical communication between computers.
2. Communication lines cuts.
3. Memory erased except for data collected prior to 2.
4. Input into each computer.

4.5. Metered rounds of communication ← Repeat

5. Output by each computer.

Build up table of input-output correlations

What correlations can be produced using ~~$X$ amount~~ 1 bit of communication?

# Bell Inequalities with Auxiliary Communication

Bell Inequalities: what correlations can be simulated with shared randomness and no communication.

$$|\langle \mathbf{A}_1\mathbf{B}_1\rangle + \langle \mathbf{A}_1\mathbf{B}_2\rangle + \langle \mathbf{A}_2\mathbf{B}_1\rangle - \langle \mathbf{A}_1\mathbf{B}_2\rangle| \leq 2$$

Bell Inequalities with Auxiliary Communication: what correlations can be simulated with shared randomness and bounded communication

$$|-\langle \mathbf{A}_1\mathbf{B}_2\rangle + \langle \mathbf{A}_1\mathbf{B}_3\rangle - \langle \mathbf{A}_2\mathbf{B}_1\rangle + \langle \mathbf{A}_2\mathbf{B}_2\rangle + \langle \mathbf{A}_2\mathbf{B}_2\rangle + \langle \mathbf{A}_3\mathbf{B}_1\rangle + \langle \mathbf{A}_3\mathbf{B}_2\rangle + \langle \mathbf{A}_3\mathbf{B}_3\rangle| \leq 6$$

1 bit inequality

**Quantify nonlocality!**





"Bell Inequalities with Auxiliary Communication", D. **Bacon** and B. F. **Toner**, quant-ph 0208057

# Simulating Measurements On a Bell Pair



ALICE

$\hat{a}$

$A \in \{+1, -1\}$

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$\langle A \rangle = \langle B \rangle = 0$$

$$\langle AB \rangle = -\hat{a} \cdot \hat{b}$$

BOB

$\hat{b}$



$B \in \{+1, -1\}$

Sending $\hat{a}$ from Alice to Bob requires $\infty$ bits of communication

Protocol for simulating Bell pair quantum correlations using (1) shared randomness and (2) a single bit of communication!

**"Correlations on Bell pair" =**
**"shared randomness" + "1 bit"**



Bacon



Toner

# More IQI Entanglement Work

**Entanglement of purification** (Terhal, Horodecki, Leung, DiVincenzo)
measure of both quantum and classical correlations

**First calculation of entanglement cost** (Vidal, Dur, Cirac)
additivity of entanglement of formation for some states

**Continuity of asymptotic entanglement measures** (Vidal)
general property of asymptotic entanglement measures

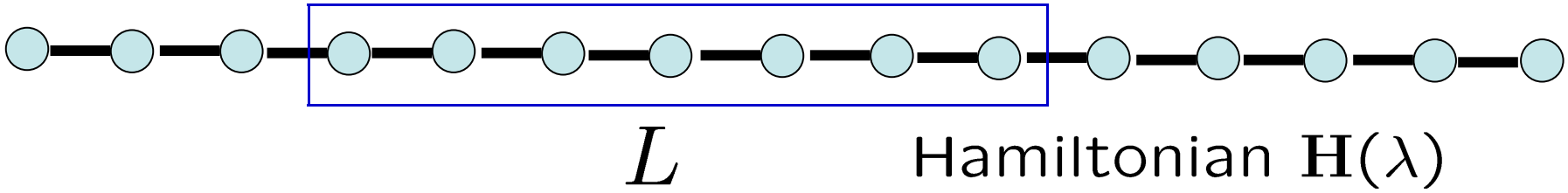
**Entanglement entropy for multipartite pure states** (Bravyi)
multipartite entropy leading to near-maximally entangled multipartite states

**Generating set for tripartite state** (Acin, Vidal, Cirac)
copies of Bell pair and GHZ states is not enough to assymptotically produce all tripartite states.

**Entanglement in symmetric states** (Stockton, Geremia, Doherty, Mabuchi)
entanglement in many-qubit symmetric states

**Local transformations of three-qubit pure state** (Spedalieri)

# Entanglement in Quantum Critical Phenomena



$L$ Hamiltonian $\mathbf{H}(\lambda)$

Quantum phase transition as of $\lambda$

Entropy when traced down to $L$ spins

noncritical: Entropy saturates or vanishes

critical: Entropy scales logarithmically with $L$

Entanglement related to breakdown of density matrix renormalization group methods at critical point

"Entanglement in quantum critical phenomena", G. Vidal, J.I. Latorre, E. Rico, and A. Kitaev, quant-ph 0211074

# Entanglement Directions

Can we show all entangled states violate local realism or converse?

Does “little entanglement” mean classically easily simulatable?

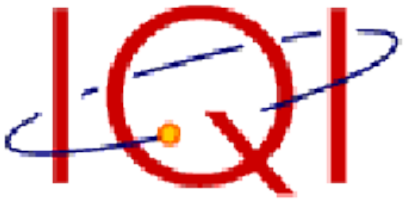
How can we understand why entanglement achieves higher winning probabilities for certain games?

Does there exist “The Game” from which one can derive quantum theory as the best strategy for this game?

Does multipartite entanglement theory lead to better algorithms for
simulating condensed matter systems?

What is does theory of entanglement look like for quantum gravity where the notion of locality is not a priori given?

# Brought To You By



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